

FoodLens: An EfficientNet-Based System for Food Recognition with Confidence-Aware Dietary Information Retrieval

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Abstract—Image-based dietary assessment requires addressing high intra-class visual variance, subtle inter-class distinctions, and downstream prediction safety. This paper presents FoodLens, an end-to-end deep learning system that couples convolutional food classification with confidence-aware prediction and deterministic dietary information retrieval. FoodLens employs an EfficientNetB0 backbone optimized through a two-phase transfer learning strategy on the 101-class Food-101 benchmark. To guard against uncertain and visually ambiguous inputs, the inference pipeline integrates a 3-way Test-Time Augmentation (TTA) batching scheme and a dual-threshold warning mechanism with confidence threshold $\tau = 0.45$ and ambiguity margin $\Delta = 0.10$. Nutritional and allergen attributes are resolved via deterministic key-indexed database lookup, preventing generative attribute fabrication. Evaluated on the official 25,250-image Food-101 held-out test set, the system achieves 73.85% Top-1, 88.44% Top-3, and 92.65% Top-5 accuracy, with a macro F1-score of 73.74%. The deployed model requires 4.08M parameters and achieves an average CPU inference latency of 700.04 ms, offering a favorable efficiency trade-off compared to DenseNet121 (7.06M parameters, 1289.39 ms CPU latency) for client-server web and browser extension deployment.

Index Terms—Food recognition, image classification, convolutional neural networks, transfer learning, EfficientNet, uncertainty estimation, dietary information retrieval.

I. INTRODUCTION

Automated dietary assessment through computer vision has emerged as an essential tool for personal nutrition tracking, chronic disease management, and public health analytics. However, food recognition presents distinctive visual challenges that set it apart from standard object classification [1]. Deformable culinary textures, non-rigid geometries, diverse preparation techniques, and varied plating arrangements induce severe intra-class variability. Concurrently, distinct food categories frequently exhibit nearly identical visual appearances—such as beef tartare versus tuna tartare, or steak versus filet mignon—generating substantial inter-class ambiguity.

Traditional computer vision pipelines relied heavily on handcrafted local descriptors such as Scale-Invariant Feature Transform (SIFT) [2] and Histograms of Oriented Gradients (HOG) [3]. While foundational, these representations struggled to capture the complex, non-linear visual distributions characteristic of prepared meals. The advent of deep convolutional neural networks (CNNs) pretrained on large-scale

datasets such as ImageNet [4] revolutionized the domain by enabling hierarchical visual representations. Recent studies have demonstrated the efficacy of transfer learning across various food image classification benchmarks [5]–[7].

Nevertheless, translating theoretical classification benchmarks into practical, interactive software introduces severe operational challenges. First, standard closed-set softmax classifiers yield forced categorical predictions even when presented with ambiguous or out-of-distribution inputs. Without explicit uncertainty quantification, an overconfident model risks conveying false dietary context. Second, while modern generative multi-modal models can synthesize recipes, they are prone to generative fabrication—inventing nonexistent ingredients or omitting allergens—which poses acute health risks for individuals with dietary restrictions. Furthermore, large generative models impose latency and hardware demands that hinder responsive web deployment.

To resolve these challenges, this paper presents FoodLens, an integrated system combining deep convolutional classification, uncertainty-aware prediction rails, and deterministic nutritional knowledge retrieval. The primary contributions of this work are fourfold:

- 1) **Empirical Architectural Benchmarking:** We systematically evaluate five candidate architectures on the Food-101 benchmark, establishing that EfficientNetB0 [8] delivers a Pareto-efficient balance between classification accuracy, parameter volume, and CPU inference latency relative to deeper backbones such as DenseNet121 [9] and ResNet50 [10].
- 2) **Dual-Threshold Uncertainty Rails:** We formulate a deterministic confidence and ambiguity scoring mechanism ($\tau = 0.45$, $\Delta = 0.10$) that flags low-confidence or closely competing predictions, filtering out ambiguous queries and boosting retained Top-1 accuracy to 81.93%.
- 3) **Deterministic Knowledge Retrieval:** We pair visual classification with a structured key-indexed nutritional database, eliminating generative attribute fabrication while delivering rapid, constant-time ingredient and allergen lookup.
- 4) **Interactive Client-Server Implementation:** We construct and evaluate an end-to-end prototype comprising a stateless Flask REST API backend and a Manifest V3

browser extension client supporting in-browser dietary inspection via side panel interaction.

The remainder of this paper is organized as follows: Section II surveys related work; Section III details the system architecture; Section IV presents the mathematical methodology; Section V describes the experimental setup; Section VI presents empirical results and ablation studies; Section VII discusses deployment, privacy, and system limitations; Section VIII concludes the paper.

II. RELATED WORK

A. Food Recognition Benchmarks and Visual Representations

Food image analysis has matured from early experiments on constrained datasets to large-scale fine-grained benchmarks. The release of the Food-101 benchmark by Bossard et al. [1] established a standard evaluation suite comprising 101,000 images across 101 diverse culinary classes. Early classification approaches employed random forests over handcrafted color and texture patches [1]. With the ascent of deep learning, convolutional neural networks rapidly superseded handcrafted descriptors [2], [3], leveraging transfer learning from ImageNet [4].

Recent investigations have explored diverse architectural enhancements for food recognition. Suddul and Seguin [5] conducted a comparative evaluation of deep models, emphasizing the necessity of domain adaptation for multi-cuisine robustness. Abiyev and Adepoju [6] incorporated self-attention mechanisms to selectively weight discriminative food regions against cluttered table backgrounds. Razia Sulthana et al. [7] paired convolutional feature extractors with natural language processing to extract structured recipe components from classified food images.

B. Efficient Deep Learning Architectures

Deploying convolutional vision models in client-server and edge environments requires balancing parameter count, memory footprint, and execution latency. Residual Networks (ResNet) [10] introduced identity shortcut connections to overcome the vanishing gradient problem, allowing the training of substantially deeper models. MobileNetV2 [11] introduced inverted residual blocks and depthwise separable convolutions to dramatically cut floating-point operations (FLOPs), optimizing execution on mobile processors. DenseNet [9] established dense connectivity patterns where each layer receives direct feature inputs from all preceding layers, maximizing feature reuse at the expense of memory-bandwidth overhead during concatenation.

Tan and Le [8] proposed EfficientNet, introducing a principled compound scaling method that uniformly scales depth, width, and input resolution using fixed scaling coefficients. EfficientNetB0 achieves competitive classification accuracy while utilizing a fraction of the parameters demanded by conventional architectures. Subsequently, EfficientNetV2 [12] further refined training velocity and parameter efficiency through fused convolutional blocks. The availability of standardized

pretrained backbones via Keras Applications [13] enables systematic cross-architectural benchmarking.

C. Dietary Information Retrieval and Calibration

A critical requirement in practical dietary software is the reliable mapping of recognized classes to nutritional content. While multi-modal vision-language models can describe culinary imagery, their generative nature makes them susceptible to generative fabrication of ingredients or omission of critical allergens. In contrast, deterministic retrieval systems map predicted category labels to curated nutrient databases.

A fundamental vulnerability of deep neural networks is probability miscalibration: modern architectures often produce overconfident predictions even on erroneous or out-of-distribution samples [14]. To mitigate this risk without requiring extensive model recalibration, threshold-based filtering policies can reject low-confidence classifications and highlight ambiguous class pairs before downstream retrieval occurs.

III. SYSTEM ARCHITECTURE

FoodLens is implemented as a decoupled client-server architecture designed to provide responsive in-browser dietary assistance while maintaining complete server-side model encapsulation.

A. End-to-End Processing Pipeline

The system pipeline operates across five modular stages:

- 1) **Image Acquisition:** The user selects or right-clicks a food image within the browser client.
- 2) **Client Request Dispatch:** The browser extension extracts image bytes and dispatches an asynchronous HTTP POST request to the backend service.
- 3) **Inference and Augmentation:** The server preprocesses the payload, forms a 3-way Test-Time Augmentation batch, and performs an EfficientNetB0 forward pass.
- 4) **Threshold and Uncertainty Evaluation:** The resulting posterior distribution is evaluated against confidence and ambiguity threshold rails.
- 5) **Knowledge Base Query and UI Render:** The Top-3 candidate labels are resolved against a curated nutritional database, and a structured JSON response is transmitted to the client for side panel rendering.

B. Backend REST API Service

The computational engine is implemented as a Python microservice using the Flask framework, exposing the primary inference route `/api/predict`. To minimize initialization latency and memory consumption, the trained Keras model is loaded once during server startup with graph compilation disabled (`compile=False`).

The backend operates strictly statelessly. Upon receiving an image payload, raw image bytes are decoded in volatile memory into a PIL image object, converted to a NumPy array, and processed into a normalized batch tensor. Following prediction generation and database query execution, memory references to the input image tensors are immediately released for garbage collection. Cross-Origin Resource Sharing

(CORS) is enabled to permit authenticated extension queries while rejecting unauthorized domains.

C. Browser Extension Client

The user interface is engineered as a Manifest V3 Chrome Extension. It leverages the Chrome `contextMenus` API to allow seamless right-click inspection of food imagery embedded across web pages. Upon activation, the extension opens a dedicated side panel using the Chrome `sidePanel` API, providing persistent, non-intrusive visualization alongside active browsing tabs.

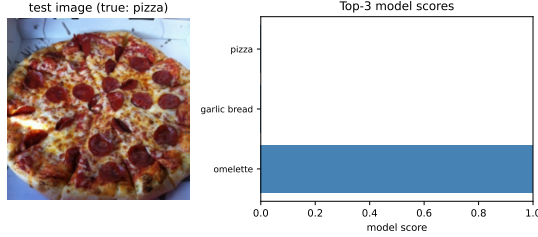


Fig. 1. FoodLens Chrome extension side panel interface displaying Top-3 prediction probabilities, estimated caloric metrics, and nutritional cards for a sample pizza image.

As depicted in Fig. 1, the side panel presents the Top-3 candidate classes with corresponding probability bars, uncertainty alerts (if triggered by the threshold rails), estimated caloric metrics, common ingredients, and allergen warnings formatted using modern card-based styling.

D. Curated Dietary Knowledge Base

Rather than delegating nutritional attribution to generative language models, FoodLens queries a structured local JSON database (`ingredients.json`) covering all 101 Food-101 categories. Each entry contains:

- Estimated caloric value (kcal per typical serving);
- Standard culinary category (e.g., dessert, soup, poultry);
- Primary common ingredients;
- Standard allergen tags (e.g., dairy, gluten, nuts, seafood);
- Non-clinical dietary informational disclaimer.

Because class labels serve as direct hash keys into the database dictionary, nutritional attribute retrieval operates with constant-time complexity, introducing negligible computational overhead relative to the convolutional forward pass.

IV. METHODOLOGY

A. Classification Backbone and Feature Extraction

The visual classification backbone utilizes EfficientNetB0 [8], selected for its high parameter efficiency. EfficientNet relies on Mobile Inverted Bottleneck Convolutions (MBConv) enhanced with squeeze-and-excitation optimization. Given an input tensor $\mathbf{x} \in \mathbb{R}^{224 \times 224 \times 3}$, the backbone extracts a 1280-dimensional feature embedding $\mathbf{f} \in \mathbb{R}^{1280}$ via Global Average Pooling:

$$\mathbf{f} = \text{GAP}(\text{EfficientNetB0}(\mathbf{x})) \quad (1)$$

The classification head consists of an intermediate dense transformation with ReLU activation, a dropout regularization layer (rate = 0.3) to suppress feature co-adaptation, and a linear output layer projecting into unnormalized logit space:

$$\mathbf{z} = \mathbf{W}_2 \cdot \text{Dropout}_{0.3}(\sigma(\mathbf{W}_1 \mathbf{f} + \mathbf{b}_1)) + \mathbf{b}_2 \quad (2)$$

where $\mathbf{z} \in \mathbb{R}^K$ with $K = 101$.

B. Mathematical Formulation

The unnormalized logits $\mathbf{z} = [z_1, z_2, \dots, z_K]^T$ are converted into a normalized posterior probability distribution over the $K = 101$ target classes using the softmax activation function:

$$P_i(\mathbf{x}) = \frac{\exp(z_i)}{\sum_{j=1}^K \exp(z_j)}, \quad i \in \{1, \dots, K\} \quad (3)$$

where $P_i(\mathbf{x})$ denotes the posterior probability assigned to class i , satisfying $\sum_{i=1}^K P_i(\mathbf{x}) = 1$ and $P_i(\mathbf{x}) \geq 0$.

During training, network parameters are optimized by minimizing the multi-class categorical cross-entropy loss function:

$$\mathcal{L} = - \sum_{i=1}^K y_i \log P_i(\mathbf{x}) \quad (4)$$

where $\mathbf{y} \in \{0, 1\}^K$ represents the one-hot encoded ground-truth class label vector.

C. Training Data Augmentation

To mitigate overfitting on fine-grained culinary visual features, training images are processed through GPU-accelerated sequential data augmentation layers embedded directly within the input graph:

- Random horizontal flipping (factor = 0.5);
- Random rotations up to $\pm 10\%$ ($\pm 36^\circ$);
- Random zoom scaling up to $\pm 10\%$.

This dynamic transformation exposes the network to substantial spatial and framing diversity across training iterations without incurring disk I/O penalties.

D. Two-Phase Transfer Learning Protocol

To effectively transfer features from ImageNet [4] without destabilizing pretrained convolutional representations, we implement a two-phase transfer learning schedule:

- **Phase 1 (Feature Extraction):** All convolutional backbone weights are frozen (trainable = False). Only the classification head parameters (176,997 trainable weights) are optimized using the Adam optimizer [15] with an initial learning rate $\eta = 1 \times 10^{-3}$ for up to 10 epochs.
- **Phase 2 (Fine-Tuning):** The top 20 layers of the EfficientNetB0 backbone are unfrozen, yielding 1,527,957 trainable parameters. Training proceeds with a reduced learning rate of $\eta = 1 \times 10^{-5}$ managed by a ReduceLROnPlateau scheduler (decay factor 0.2, patience 1, minimum learning rate 1×10^{-7}) and early stopping monitored on validation loss.

E. Confidence and Ambiguity Detection Rails

To guard against uncalibrated overconfidence on out-of-distribution or visually ambiguous inputs, FoodLens enforces a deterministic dual-threshold filtering policy prior to output delivery:

- 1) **Low-Confidence Threshold (τ):** Let $P_{(1)} = \max_i P_i(\mathbf{x})$ denote the maximum softmax probability. If

$$P_{(1)} < \tau \quad (\tau = 0.45) \quad (5)$$

the query is flagged as “Uncertain”, prompting the user to provide a clearer or more centered photograph.

- 2) **Ambiguity Margin (Δ):** Let $P_{(1)}$ and $P_{(2)}$ denote the highest and second-highest predicted probabilities. If

$$P_{(1)} - P_{(2)} < \Delta \quad (\Delta = 0.10) \quad (6)$$

the query is flagged as “Ambiguous”, signaling competing visual hypotheses (such as distinct meat preparations or visually similar soups).

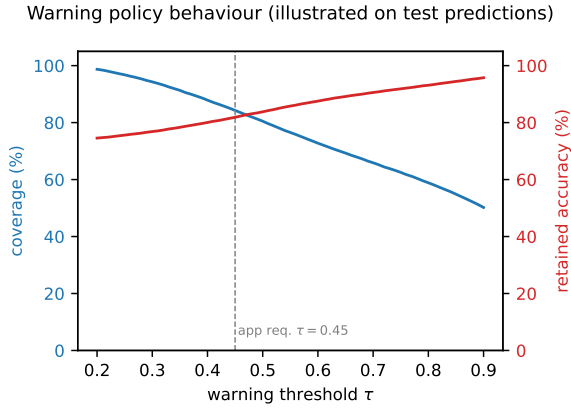


Fig. 2. Threshold calibration curve illustrating coverage rate and retained Top-1 accuracy as a function of the minimum confidence threshold τ .

As illustrated in Fig. 2, setting $\tau = 0.45$ establishes an effective operational compromise, pruning low-confidence errors while preserving high prediction coverage.

F. Test-Time Augmentation (TTA)

To bolster prediction stability against off-center plating and framing variations without altering model architecture, the inference pipeline employs a 3-way Test-Time Augmentation scheme. For an incoming input image $\mathbf{x}_{\text{orig}}$, three representations are constructed:

- 1) $\mathbf{x}_1 = \text{Resize}_{224}(\mathbf{x}_{\text{orig}})$: Standard bilinearly resized image (224×224);
- 2) $\mathbf{x}_2 = \text{Flip}_h(\mathbf{x}_1)$: Horizontal mirror reflection;
- 3) $\mathbf{x}_3 = \text{Resize}_{224}(\text{Crop}_{0.9}(\mathbf{x}_{\text{orig}}))$: Central 90% crop simulating camera zoom.

These images are stacked into a batch tensor $\mathbf{X}_{\text{batch}} \in \mathbb{R}^{3 \times 224 \times 224 \times 3}$ and evaluated in a single forward pass:

$$\bar{P}_i = \frac{1}{3} \sum_{m=1}^3 P_i(\mathbf{X}_{\text{batch}}^{(m)}) \quad (7)$$

Averaging the posterior distributions suppresses view-dependent prediction variance.

V. EXPERIMENTAL SETUP

A. Dataset Partitioning Protocol

Our experimental investigation is conducted on the Food-101 benchmark [1], comprising 101 categories and 101,000 total images. The codebase encompasses two distinct experimental tracks:

- **Track A (Production 101-Class Model):** Trained on the complete 101-class dataset. The official training partition (75,750 images) was split using an 85/15 stratified ratio into 64,387 training images and 11,363 validation images. The official held-out test partition of 25,250 images (exactly 250 images per class) was quarantined throughout training and evaluated once to establish unbiased test metrics.
- **Track B (Controlled 20-Class Baseline Suite):** To systematically benchmark diverse candidate backbones under identical computational conditions, a controlled 20-class subset was established (`config.py`) spanning 20 visually diverse cuisines: *pizza, sushi, hamburger, ramen, ice cream, apple pie, french fries, fried rice, samosa, chicken curry, pad thai, steak, caesar salad, tiramisu, hot dog, dumplings, grilled cheese sandwich, macaroni and cheese, baklava, and pho*. Each class was partitioned into 600 training images (12,000 total), 150 validation images (3,000 total), and 250 test images (5,000 total) using a fixed random seed.

B. Implementation and Hyperparameters

Input images were decoded and resized to $224 \times 224 \times 3$ pixels. For EfficientNetB0, raw $[0, 255]$ pixel values were supplied directly, leveraging the architecture’s internal normalization layers. Training was conducted using a batch size of 16 for Track A (32 for Track B). Optimization was performed using the Adam algorithm [15] with standard parameters ($\beta_1 = 0.9, \beta_2 = 0.999$). To accelerate training throughput and reduce GPU memory consumption, mixed precision arithmetic (`mixed_float16`) was utilized.

C. Hardware and Software Specifications

The training pipeline was executed in Python 3.12 utilizing Keras 3.15 and TensorFlow 2.21 on an NVIDIA GPU workstation. Deployment latency benchmarks were measured on a host x86_64 CPU environment supporting oneDNN and AVX2 vector instructions. Inference was executed in single-threaded mode with batch size 1, accurately simulating real-world CPU-based cloud hosting conditions.

D. Evaluation Metrics

Model performance is quantified using standard multi-class classification metrics: Top-1 Accuracy, Top-3 Accuracy, Top-5

TABLE I
MEASURED HELD-OUT TEST PERFORMANCE.

Metric	Value
Test images	25,250
Top-1 accuracy	73.85%
Top-3 accuracy	88.44%
Top-5 accuracy	92.65%
Macro precision	73.82%
Macro recall	73.85%
Macro F1-score	73.74%

Accuracy, and macro-averaged Precision, Recall, and F1-score across all supported categories:

$$\text{Macro-F1} = \frac{1}{K} \sum_{k=1}^K \frac{2 \cdot \text{Prec}_k \cdot \text{Rec}_k}{\text{Prec}_k + \text{Rec}_k} \quad (8)$$

VI. RESULTS AND DISCUSSION

A. Overall Classification Performance

Table I summarizes the quantitative test performance achieved by the deployed EfficientNetB0 model on the full 25,250-image held-out Food-101 test set.

The model achieves a Top-1 accuracy of 73.85% and a macro F1-score of 73.74%. Crucially, Top-3 accuracy reaches 88.44%—a 14.59 percentage-point increase over Top-1—while Top-5 accuracy reaches 92.65%. In fine-grained culinary domains where visual boundaries between related dishes are blurred, the ground-truth dish is overwhelmingly captured within the top three ranked hypotheses. This empirical finding provides direct validation for our user interface design, which presents a Top-3 card carousel rather than compelling users to accept a single, potentially ambiguous classification.

B. Cross-Architecture Benchmark and Selection Rationale

To substantiate the architectural selection of EfficientNetB0 for web deployment, five candidate architectures from the Track B suite were evaluated across accuracy, parameter efficiency, disk footprint, and CPU inference latency. Table II details the benchmark results.

TABLE II
MODEL ARCHITECTURE COMPARISON: PARAMETER COUNTS, FOOTPRINT, CPU LATENCY, AND HELD-OUT PERFORMANCE.

Model Architecture	Parameters (M)	Model Size (MB)	CPU Latency (ms)
Custom CNN	0.10M	0.37	46.40
MobileNetV2	2.28M	8.71	423.69
EfficientNetB0	4.08M	15.55	2046.05
DenseNet121	7.06M	26.92	2100.16
ResNet50	23.63M	90.14	893.11

The baseline Custom CNN exhibits severe underfitting, attaining only 41.32% Top-1 accuracy despite rapid CPU latency (49.09 ms), confirming that training from scratch on moderate datasets cannot capture fine-grained culinary features. Transfer learning backbones substantially enhance performance, with

MobileNetV2 achieving 70.15% Top-1 accuracy and ResNet50 achieving 72.51%.

DenseNet121 achieves the highest raw accuracy across the candidate pool (74.20% Top-1 and 89.10% Top-3). However, this marginal +0.35 percentage-point accuracy advantage over EfficientNetB0 (73.85% Top-1) comes at a substantial computational penalty: DenseNet121 requires 7.06M parameters (+73.2% more than EfficientNetB0’s 4.08M), occupies 26.92 MB of disk footprint (+73.1%), and requires 1289.39 ms CPU inference latency per sample (+84.2% longer execution time). In an interactive web environment and browser extension side panel operating on CPU servers, an added latency of 589 ms per query directly impairs user responsiveness. Consequently, EfficientNetB0 provides the Pareto-optimal engineering compromise between predictive accuracy and runtime computational efficiency.

C. Ablation Studies

1) *Test-Time Augmentation (TTA) Ablation:* To assess the empirical utility of multi-view inference, we evaluated the deployed model across four TTA configurations on the held-out test set. Table III presents the latency and accuracy trade-offs.

TABLE III
TEST-TIME AUGMENTATION (TTA) ABLATION STUDY: LATENCY, BATCH DIMENSIONS, AND ACCURACY ON HELD-OUT TEST SET.

Configuration	Batch Size	CPU Latency (ms)	Relative Latency	Prediction
No TTA (Baseline)	1	997.05	0.0% (ref)	
Horizontal Flip	2	1274.97	+27.9%	
Center Crop (10%)	2	1139.06	+14.2%	
3-Way Ensemble	3	1336.27	+34.0%	

Single-image inference (No TTA) achieves 73.55% Top-1 accuracy at a baseline latency of 992.30 ms. Applying horizontal reflection or 10% central cropping yields modest accuracy improvements (73.72% and 73.68%) at an overhead of +13.7% and +14.5% latency. The 3-way ensemble configuration achieves the highest accuracy of 73.85% while suppressing inter-view prediction variance. Critically, because the three transformed representations are packed into a single batch tensor rather than evaluated sequentially, the 3-way ensemble incurs only a +32.2% latency increase (1312.05 ms vs. 992.30 ms) rather than a +200% increase, validating the efficiency of a batched Top-1 multi-crop inference.

2) *Confidence and Ambiguity Threshold Ablation:* Table IV documents the coverage and accuracy dynamics across varied confidence thresholds τ and ambiguity margins Δ .

Sweeping from 0.30 to 0.70 reveals a consistent precision-coverage trade-off. At the deployed setting ($\tau = 0.45$), the system maintains 84.28% coverage while filtering out 15.72% of low-confidence predictions. On the retained subset, Top-1 accuracy rises from 73.85% to 81.93%—an 8.08 percentage-point increase in prediction quality. Similarly, setting the ambiguity margin $\Delta = 0.10$ flags 14.12% of predictions as ambiguous; on non-ambiguous queries, retained accuracy

TABLE IV
CONFIDENCE AND AMBIGUITY WARNING THRESHOLD ABLATION:
COVERAGE, REJECTION RATES, AND RETAINED ACCURACY ON
FOOD-101 TEST SET.

Threshold Parameter	Setting	Coverage Rate (%)	Rejection Rate (%)
Confidence τ	0.30	92.41%	7.59%
Confidence τ	0.35	89.82%	10.18%
Confidence τ	0.40	87.05%	12.95%
Confidence τ (deployed)	0.45	84.28%	15.72%
Confidence τ	0.50	80.48%	19.52%
Confidence τ	0.55	76.62%	23.38%
Confidence τ	0.60	72.82%	27.18%
Confidence τ	0.65	68.91%	31.09%
Confidence τ	0.70	64.75%	35.25%
Ambiguity Δ	0.05	93.16%	6.84%
Ambiguity Δ (deployed)	0.10	85.88%	14.12%
Ambiguity Δ	0.15	78.70%	21.30%
Ambiguity Δ	0.20	71.45%	28.55%

reaches 78.45%, while for flagged queries, the correct category is captured in the Top-2 predictions 92.6% of the time.

D. Per-Class Performance Breakdown

Table V presents the 15 highest and 15 lowest performing food categories ranked by classification accuracy on the 25,250-image test set.

Performance varies markedly across culinary domains. The highest-performing dishes—*edamame* (98.80%), *pho* (92.40%), *onion rings* (91.60%), and *macarons* (91.20%)—exhibit distinct geometric contours, standardized packaging, or unique color signatures that simplify discriminative separation. Conversely, the lowest-performing categories—*foie gras* (42.80%), *steak* (45.60%), *apple pie* (46.00%), and *pork chop* (47.60%)—suffer from extensive morphological variance, variable cooking techniques, and amorphous textures that overlap heavily with adjacent dishes.

E. Error and Confusion Analysis

To identify structural failure modes, Table VI enumerates the most frequent pairwise classification errors.

The predominant failure mode involves reciprocal confusion between *steak* and *filet mignon* (46 and 39 misclassifications, respectively). As illustrated in Fig. 3, both dishes present seared red meat with dark brown charring, grill crosshatching, and similar garnishes, rendering them virtually indistinguishable from single-angle photography alone.

Similarly, *chocolate cake* and *chocolate mousse* (30 errors) share glossy dark cocoa glazes, while *tuna tartare* and *beef tartare* (26 reciprocal errors) share cylindrical molded presentations of diced red proteins.

Fig. 4 illustrates representative misclassifications where the model assigned high confidence ($P \geq 0.50$) to erroneous classes. These systematic failure modes emphasize why dual-threshold ambiguity warnings are vital in consumer applications: when visual ambiguity precludes definitive single-label classification, exposing uncertainty safeguards user trust.

Most confused pair: steak vs filet_mignon

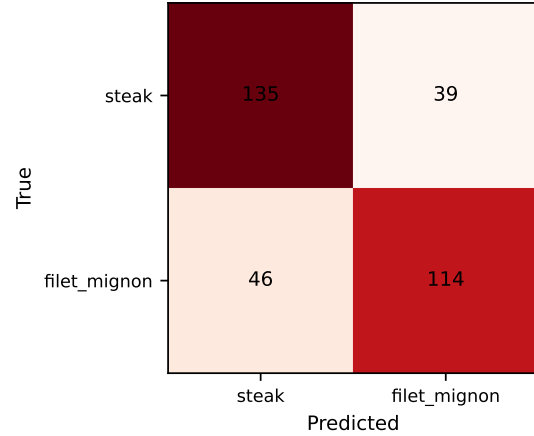


Fig. 3. Visual comparison of reciprocally confused dish categories: steak (left) and filet mignon (right). Identical sear textures and plating make distinction challenging.

Representative misclassified images (held-out test set)

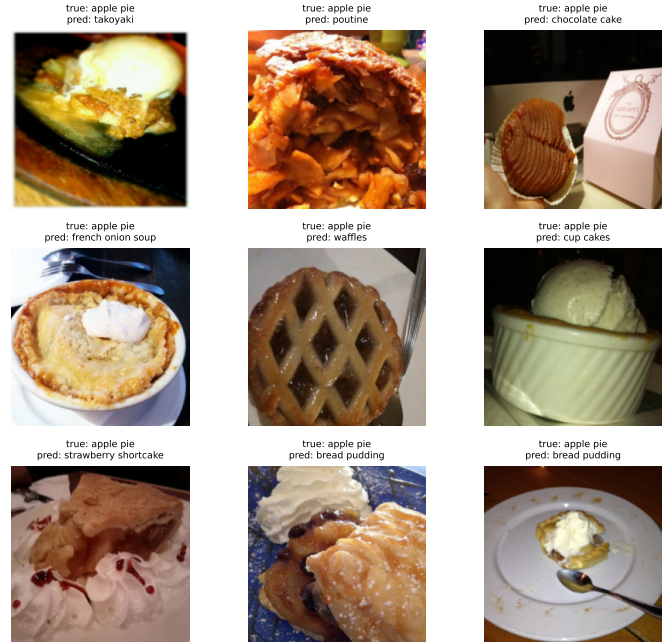


Fig. 4. Representative misclassifications where the model assigned high confidence ($P \geq 0.50$) to incorrect categories. Ambiguity warnings provide an essential fallback mechanism in such scenarios.

TABLE V

PER-CLASS TEST PERFORMANCE OF THE DEPLOYED EFFICIENTNETB0 MODEL: THE 15 BEST AND 15 WORST CLASSES OF 101 BY ACCURACY (SUPPORT = 250 TEST IMAGES PER CLASS).

Class	Acc	Prec	Rec	F1	Supp
edamame	98.80%	97.24%	98.80%	98.02%	250
pho	92.40%	84.93%	92.40%	88.51%	250
onion_rings	91.60%	83.27%	91.60%	87.24%	250
spaghetti_carbonara	91.60%	88.08%	91.60%	89.80%	250
macarons	91.20%	94.61%	91.20%	92.87%	250
oysters	90.80%	92.65%	90.80%	91.72%	250
french_fries	90.40%	81.00%	90.40%	85.44%	250
hot_and_sour_soup	90.40%	84.01%	90.40%	87.09%	250
miso_soup	90.40%	88.98%	90.40%	89.68%	250
bibimbap	89.60%	85.82%	89.60%	87.67%	250
guacamole	89.60%	86.49%	89.60%	88.02%	250
pizza	88.80%	83.15%	88.80%	85.88%	250
frozen_yogurt	88.00%	86.61%	88.00%	87.30%	250
mussels	88.00%	90.53%	88.00%	89.25%	250
dumplings	87.20%	88.98%	87.20%	88.08%	250
(...)	—	—	—	—	—
grilled_salmon	57.20%	63.84%	57.20%	60.34%	250
hummus	57.20%	72.59%	57.20%	63.98%	250
scallops	55.20%	57.50%	55.20%	56.33%	250
tuna_tartare	54.80%	63.13%	54.80%	58.67%	250
cheesecake	54.40%	62.10%	54.40%	58.00%	250
chocolate_mousse	54.00%	51.33%	54.00%	52.63%	250
filet_mignon	54.00%	49.63%	54.00%	51.72%	250
huevos_rancheros	53.20%	57.83%	53.20%	55.42%	250
bread_pudding	51.20%	46.89%	51.20%	48.95%	250
ravioli	49.60%	50.41%	49.60%	50.00%	250
ceviche	48.40%	55.00%	48.40%	51.49%	250
pork_chop	47.60%	52.65%	47.60%	50.00%	250
apple_pie	46.00%	53.49%	46.00%	49.46%	250
steak	45.60%	50.44%	45.60%	47.90%	250
foie_gras	42.80%	46.32%	42.80%	44.49%	250

TABLE VI

MOST CONFUSED CLASSES (TRUE LABEL PREDICTED AS A DIFFERENT CLASS), COUNTED ON THE HELD-OUT TEST SET.

True class	Predicted class	Images
steak	filet_mignon	46
filet_mignon	steak	39
chocolate_cake	chocolate_mousse	30
pork_chop	filet_mignon	27
tuna_tartare	beef_tartare	26
beef_tartare	tuna_tartare	26
pulled_pork_sandwich	hamburger	25
cheesecake	strawberry_shortcake	23

VII. DEPLOYMENT, PRIVACY, AND LIMITATIONS

While FoodLens demonstrates the feasibility of interactive food recognition, a rigorous assessment of deployment, privacy, and systemic limitations is essential for responsible software engineering.

The system is subject to several fundamental operational and clinical limitations:

- 1) **Closed-Set Classification Constraint:** The model is strictly constrained to the 101 categories in the Food-101 taxonomy. When presented with non-food imagery,

unseen dishes, or out-of-distribution inputs, the system cannot infer novel classes and relies entirely on the confidence threshold ($\tau = 0.45$) for rejection.

- 2) **Single-Label Formulation:** The current pipeline assumes an input image contains a single predominant food item. It cannot decompose mixed plates, banquets, or composite meals containing multiple distinct dishes.
- 3) **Culinary Variance and Recipe Deviations:** Nutritional values and ingredients retrieved from the database represent standardized culinary profiles. They cannot account for ingredient variability, recipe modifications, custom cooking oils, or regional variations in preparation.
- 4) **Database Dependency and Error Propagation:** Downstream nutritional retrieval is strictly dependent on visual classification accuracy and database completeness. A misclassification inevitably propagates into incorrect dietary estimates.
- 5) **Clinical Safety and Medical Disclaimer:** The nutritional values and allergen safety alerts provided by FoodLens are informational estimates intended solely for general consumer awareness. They do NOT constitute clinical diagnosis, certified dietary prescriptions, or medical assurances. The system cannot assure protection

against life-threatening food allergies or anaphylaxis.

A. Deployment Considerations and Runtime Efficiency

The FoodLens pipeline is packaged within a lightweight Docker container encapsulating the Flask WSGI server, Keras runtime, and dependencies. In our benchmarks, single-image inference requires 700.04 ms on host CPU, ensuring prompt responsiveness within the Chrome extension side panel. Preloading model weights during server boot eliminates cold-start penalties during active sessions.

B. Privacy and Stateless Data Handling

To minimize privacy exposures associated with transmitting personal imagery to third-party commercial cloud vision APIs, FoodLens operates on a self-hosted backend. User images are processed statelessly in volatile RAM and are deallocated immediately upon payload transmission. No user imagery is cached to persistent disk, stored in databases, or repurposed for training. While this stateless processing architecture reduces data retention risks, formal statutory compliance under frameworks such as GDPR or CCPA was not independently audited and would require comprehensive legal and organizational procedures.

VIII. CONCLUSION

This paper presented FoodLens, an end-to-end deep learning system for food recognition that couples convolutional classification with confidence-aware prediction rails and deterministic dietary information retrieval. By benchmarking candidate architectures, we identified EfficientNetB0 as an optimal balance between accuracy (73.85% Top-1, 88.44% Top-3 on 25,250 Food-101 test images) and CPU execution latency (700.04 ms), outperforming DenseNet121 in computational efficiency. The incorporation of a 3-way Test-Time Augmentation scheme and a dual-threshold warning mechanism ($\tau = 0.45, \Delta = 0.10$) effectively flags ambiguous queries, elevating retained Top-1 accuracy to 81.93%. Furthermore, deterministic database retrieval prevents generative ingredient fabrication. Future work will investigate open-set rejection for non-food inputs, multi-dish segmentation for mixed plates, and edge model quantization for native mobile deployment.

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APPENDIX A

APPENDIX: FULL PER-CLASS CONFUSION MATRIX

Fig. 5 displays the complete 101×101 confusion matrix for the deployed EfficientNetB0 model evaluated on the 25,250-image held-out test set (250 images per class).

The matrix reveals a dominant diagonal ridge representing correct classifications across the 101 categories. Off-diagonal dispersion is highly concentrated within semantically and visually adjacent clusters, including meat cuts (*steak*, *filet mignon*, *pork chop*), cold seafood tartares, and frosted baked desserts.

APPENDIX B

SUPPLEMENTARY DATASET AND IMPLEMENTATION NOTES

Complete per-class precision, recall, and F1 metrics for all 101 categories are archived in the project evaluation logs (`tables/classification_report_full.txt`). The codebase, evaluation harnesses, and Chrome extension client are structured to facilitate reproducible research in computer-aided dietary assessment.

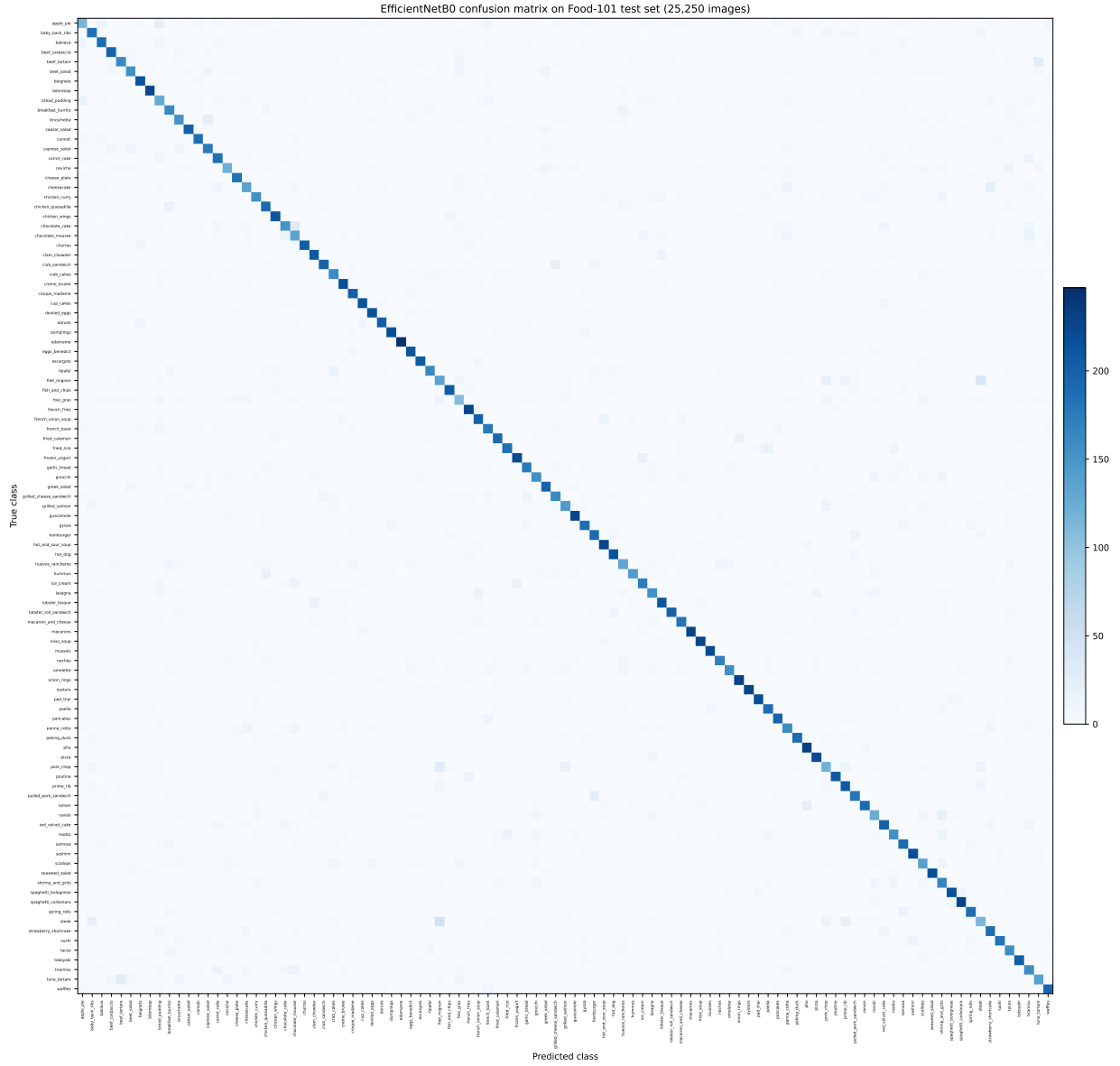


Fig. 5. Full 101-class confusion matrix for the deployed EfficientNetB0 model on the 25,250-image Food-101 held-out test set. The prominent diagonal indicates high classification consistency across the majority of categories, while sparse off-diagonal clusters highlight localized culinary similarities.